

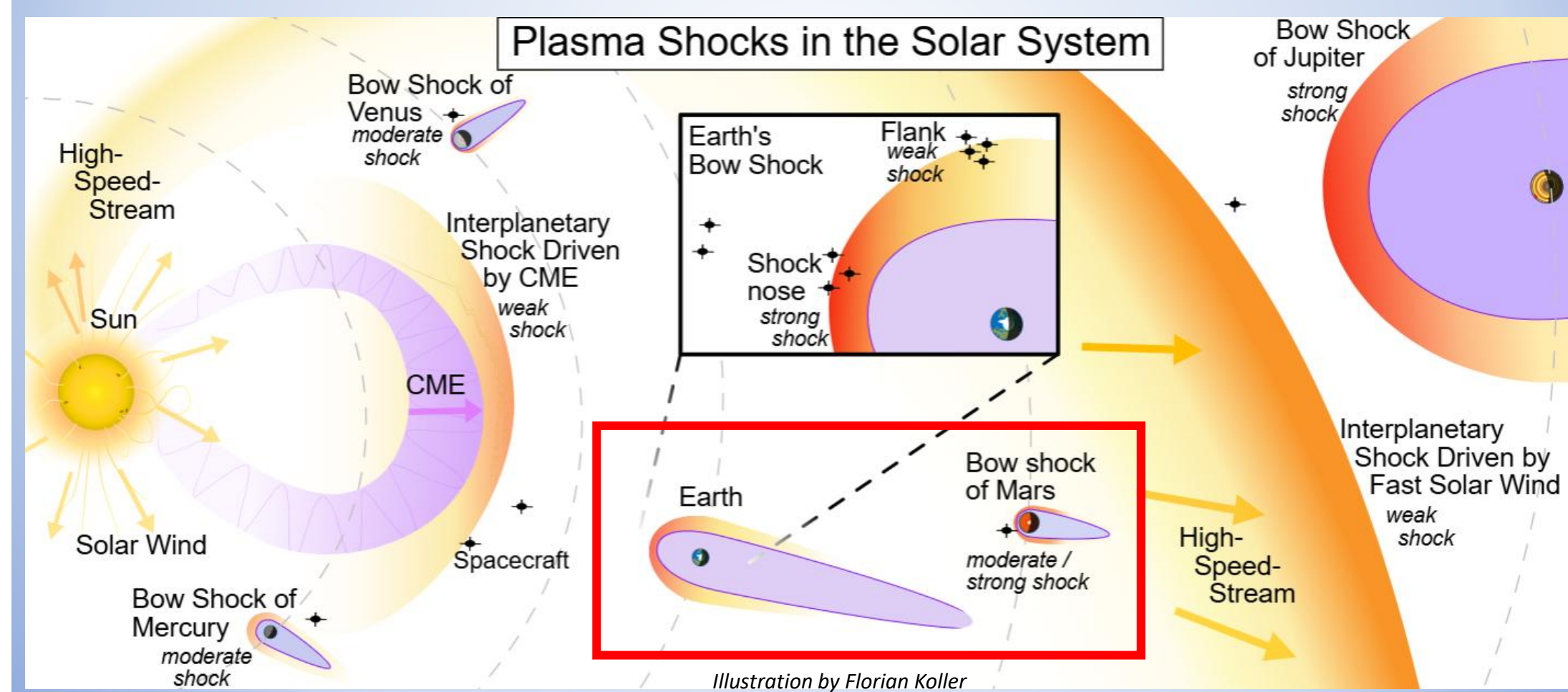
Electron acceleration comparison at heliospheric shocks

RQ1: How good are the heliospheric shocks in accelerating particles, if there are already energetic particles present upstream of the shock?

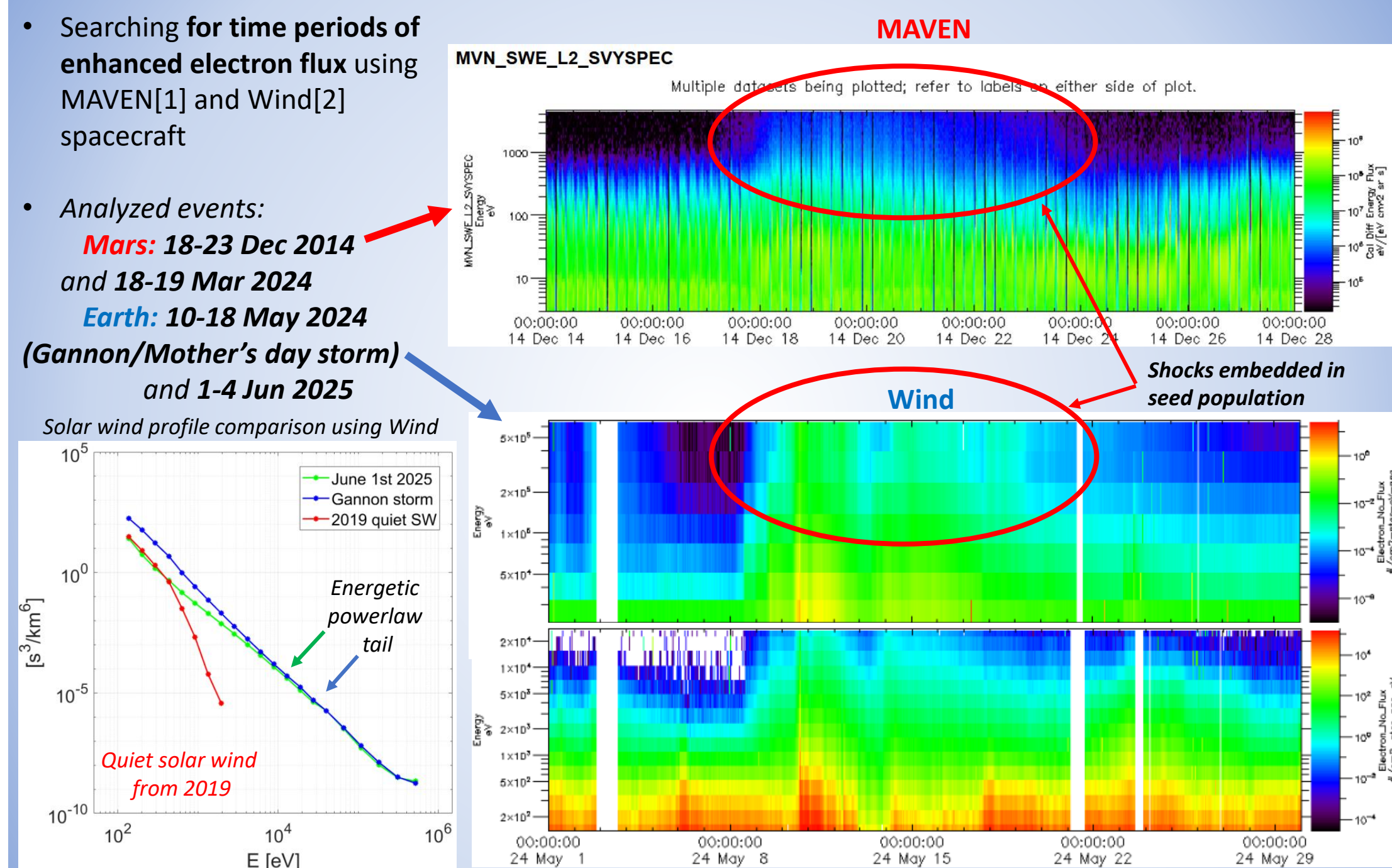
Background

RQ2: How does electron acceleration depend on bow shock size and parameters describing the shocks (θ_{Bn} , M_A , ...)

- Ongoing work!
Electron acceleration comparison of the **Martian**, and **terrestrial** quasi-perpendicular bow shocks under **seeded solar wind conditions**

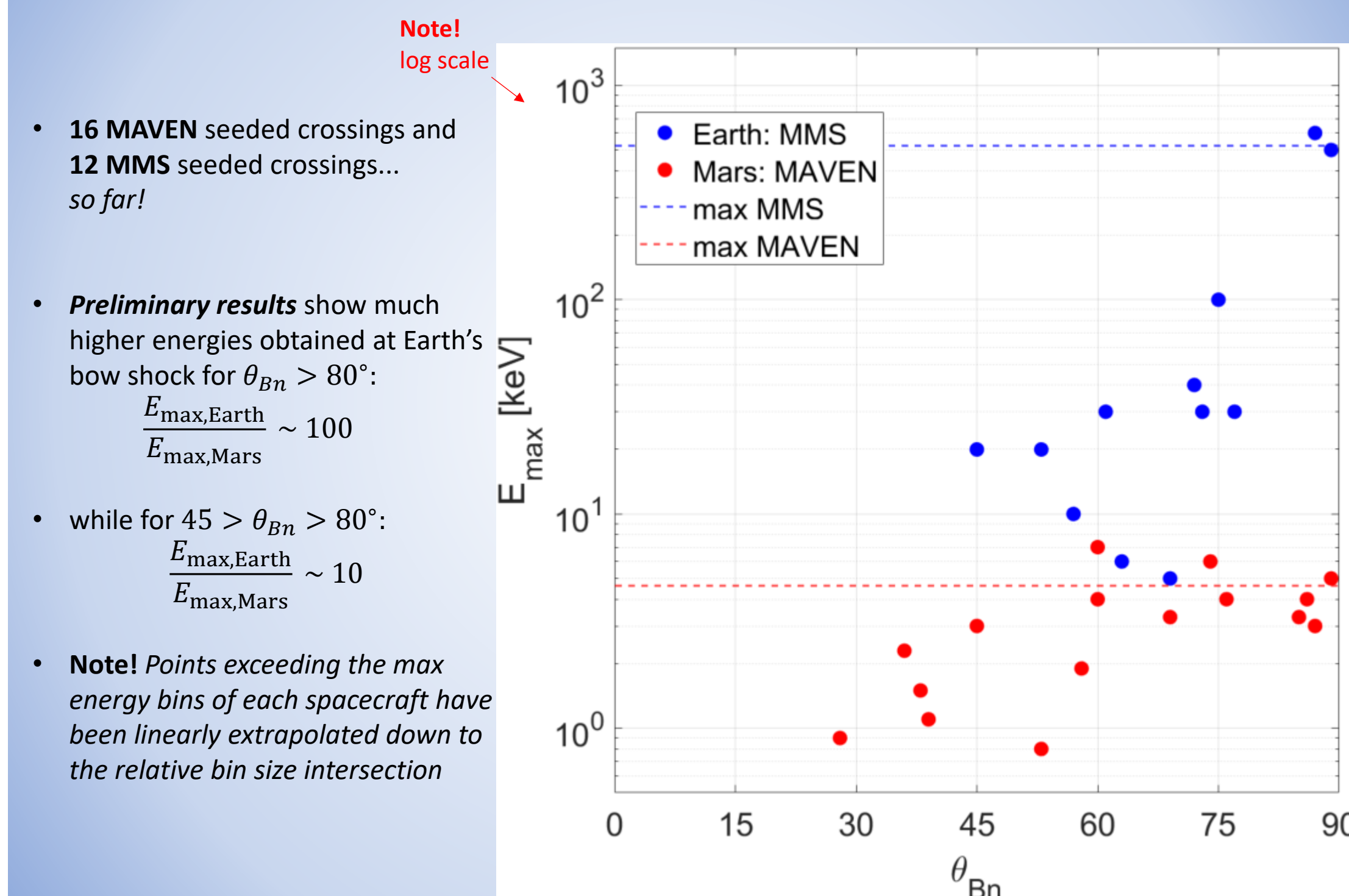


Finding Seeded Events/Shock crossings



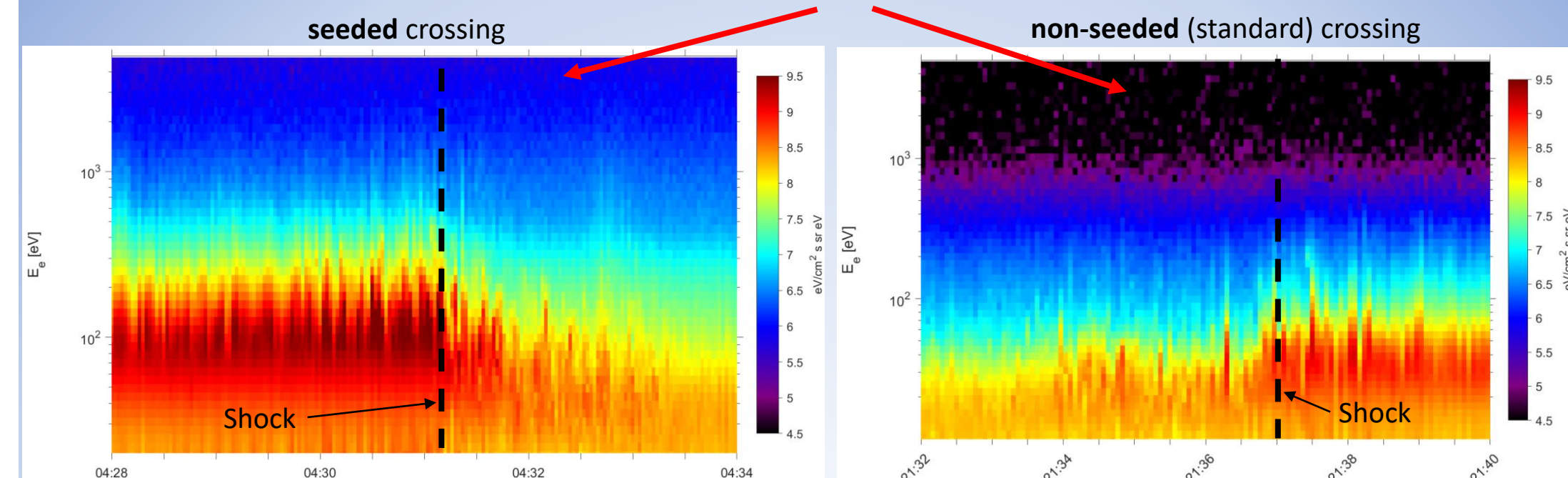
Preliminary Results

Seeded shock crossings: Mars vs Earth



Seeded shock: Example and Analysis Method

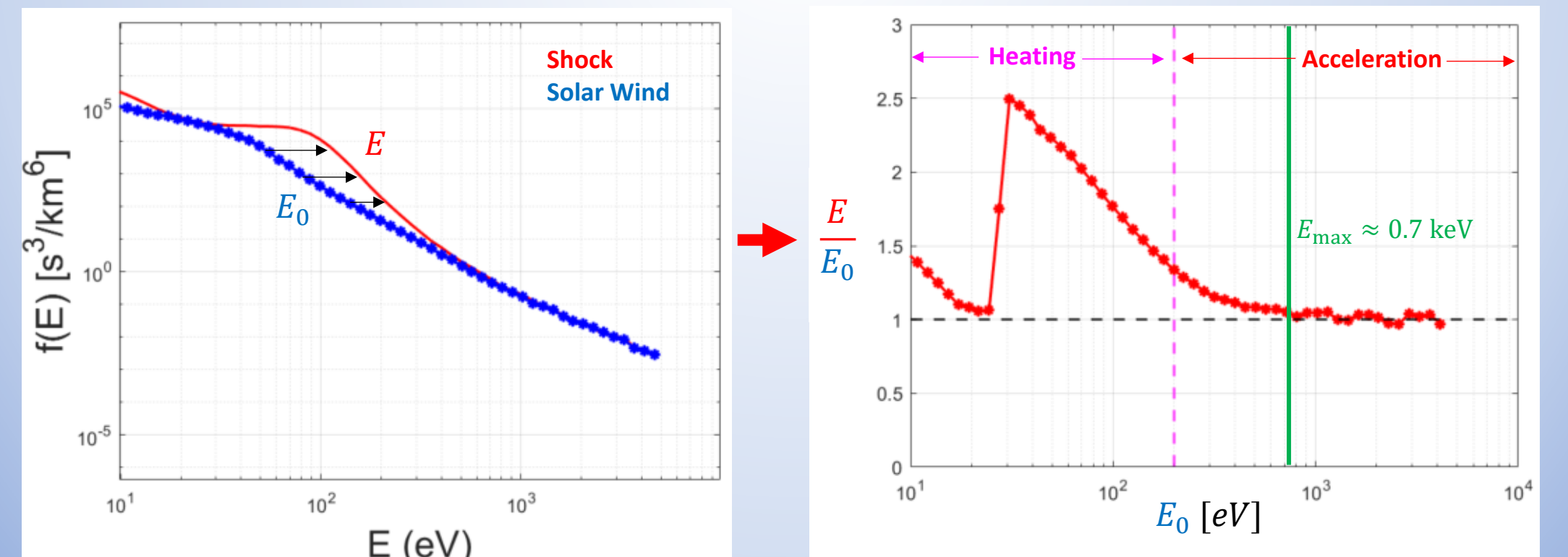
Examples from **Martian bow shock** using MAVEN:



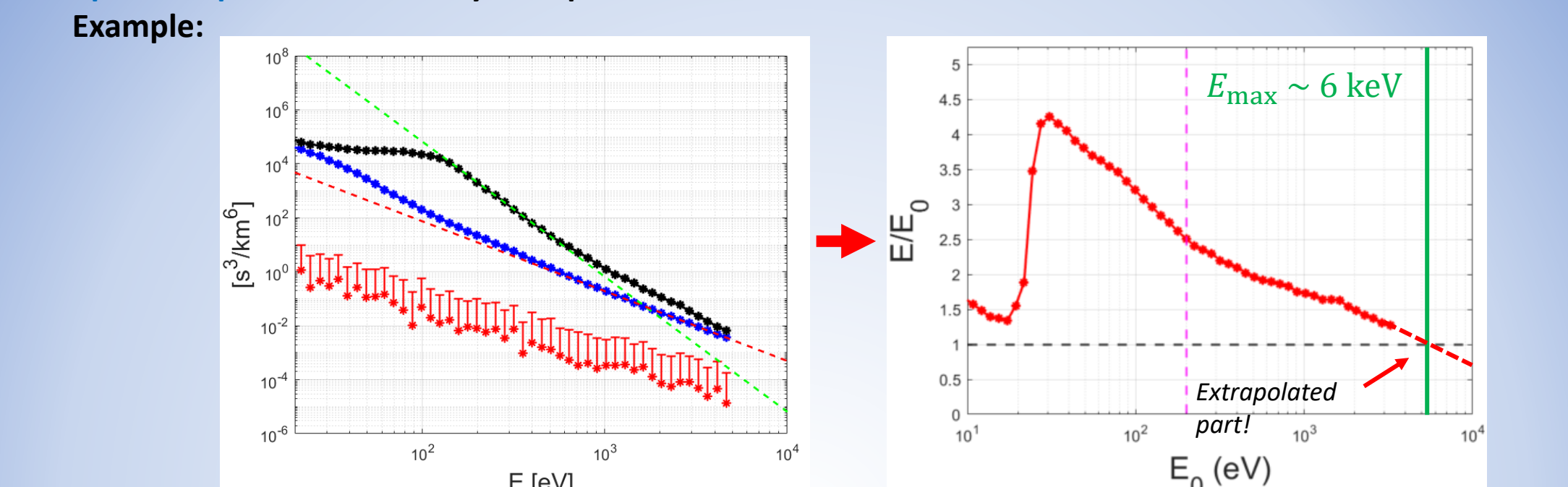
Seeded shock method: Determining $E_{\max}$

- Building on Liouville's theorem: " $f_{||}$ is constant (conserved) along the particle trajectory" *however will use **total** (omni-averaged) **distribution** $f(E)$ * (for comparison and availability purposes!)

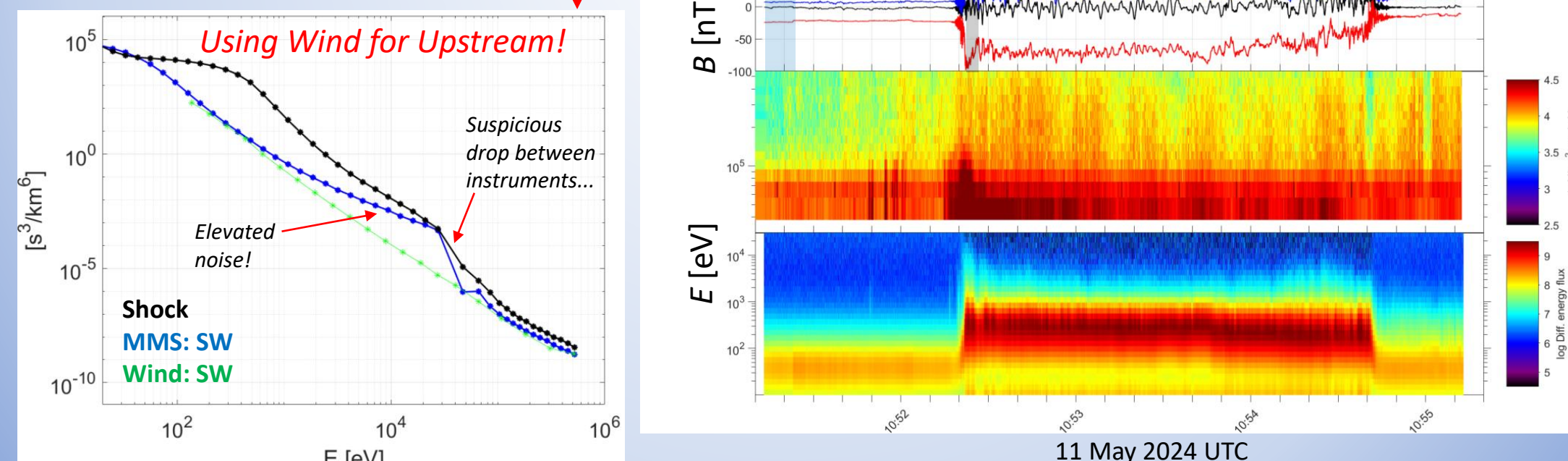
- Determine $E_{\max}$ as the energy when the **shocked distribution** (red curve) "merges" with the **upstream dist.** (blue curve) ($\equiv$ within relative energy bin size) $E_{\max} \rightarrow E/E_0 < 1.06$ MAVEN relative bin size



- Extrapolation. For the crossings where the **shocked profile** never completely merges with the **upstream profile**, we linearly extrapolate down to the relative bin size limit!



- MMS[3] issue: Instruments not designed to operate under these SW-conditions.



Conclusions and Future work

- Earth's bow shock displayed a factor of 10-100 higher energization compared to the Martian bow shock increasing with shock angle
- Search for more seeded crossings during the rest of the seeded intervals
- So far, **no quasi-parallel shocks** have been found with MMS during the seeded time period. Hence, the study can only draw conclusions for quasi perpendicular shock geometries
- As the study stands, the $E_{\max}$ determination is dependent on the energy bin size of the different spacecraft. As future work, we intend to instead use a **fixed threshold for the E/E_0** in order to treat the two datasets (Earth and Mars) on a more equal footing

Acknowledgement

Acknowledgement: The authors thank the entire MMS and MAVEN teams for the mission design and continuing operations. ML acknowledges the funding support of the Royal Society awards and the STFC grant ST/T00018X/1, RF/ERE\210353, and RF/ERE\231151. HH acknowledge the funding support from the Royal Society award URF\R1\180671.

References:

- [1] Jakosky et al. 2016, The Mars Atmosphere and Volatile Evolution (MAVEN) Mission, Space Science Reviews
- [2] Harten, R., & Clark, K. (1995). The design features of the GGS wind and polar spacecraft. Space Science Reviews
- [3] Burch et al. 2016, Magnetospheric Multiscale Overview and Science Objectives, Space Science Review,